EXPERIMENT / MODULE 4

# CLIENT-SIDE FORM VALIDATION SCRIPTING

## AIM & OBJECTIVE

To develop a comprehensive, modular JavaScript validation engine that enforces strict input constraints (rejecting numbers in alphabet fields and rejecting letters in numeric fields).

## 1. STRICT VALIDATION RULES TABLE

| Validation Rule | Target Fields | Enforcement Action & Error Display |
|---|---|---|
| Alphabet Only | Passenger Name, Cardholder Name, Username | Numbers are strictly rejected on keystroke. Displays: *'Numbers are NOT allowed!'* |
| Number Only | Mobile Number, Passenger Age, Card Number, CVV | Alphabets are strictly rejected on keystroke. Displays: *'Alphabets are NOT allowed!'* |
| Mobile Pattern | Contact Number (10 digits) | Must start with [6-9] and contain exactly 10 digits. |
| Email Pattern | User Email | Enforces regex: /^[\w.-]+@[\w.-]+\.[a-z]{2,}$/i |
| Age Limit | Passenger Age | Numeric integer between 1 and 120. |

## 2. KEY ALGORITHMIC SCRIPT IMPLEMENTATIONS

```
// 1. Alphabet-Only Validator
function validateAlphabetOnly(inputElement, fieldName) {
    const val = inputElement.value.trim();
    if (/\d/.test(val)) {
        return showError(inputElement, "Numbers are NOT allowed in " + fieldName + "! Only alphab
ets permitted.");
    }
    if (!/^[A-Za-z\s]+$/.test(val)) {
        return showError(inputElement, "Special characters NOT allowed! Only alphabets permitted.
");
    }
    return clearError(inputElement);
}

// 2. Number-Only Validator
function validateNumberOnly(inputElement, fieldName, min, max) {
    const val = inputElement.value.trim();
    if (/[a-zA-Z]/.test(val)) {
        return showError(inputElement, "Alphabets are NOT allowed in " + fieldName + "! Only numb
ers permitted.");
    }
    if (!/^[0-9]+$/.test(val)) {
        return showError(inputElement, "Special characters/spaces NOT allowed! Only numbers permi
tted.");
    }
    return clearError(inputElement);
}
```

## 3. VERIFICATION & TEST RESULTS

| Input Tested | Field | Expected Result | Observed Status |
|---|---|---|---|
| Manikandan123 | Name | Reject digits 123, show error | **PASS** (Digits blocked) |
| 98765abcde | Mobile | Reject letters abcde, show error | **PASS** (Letters blocked) |
| 35 Yrs | Age | Valid numeric value 35 allowed | **PASS** (Accepted) |